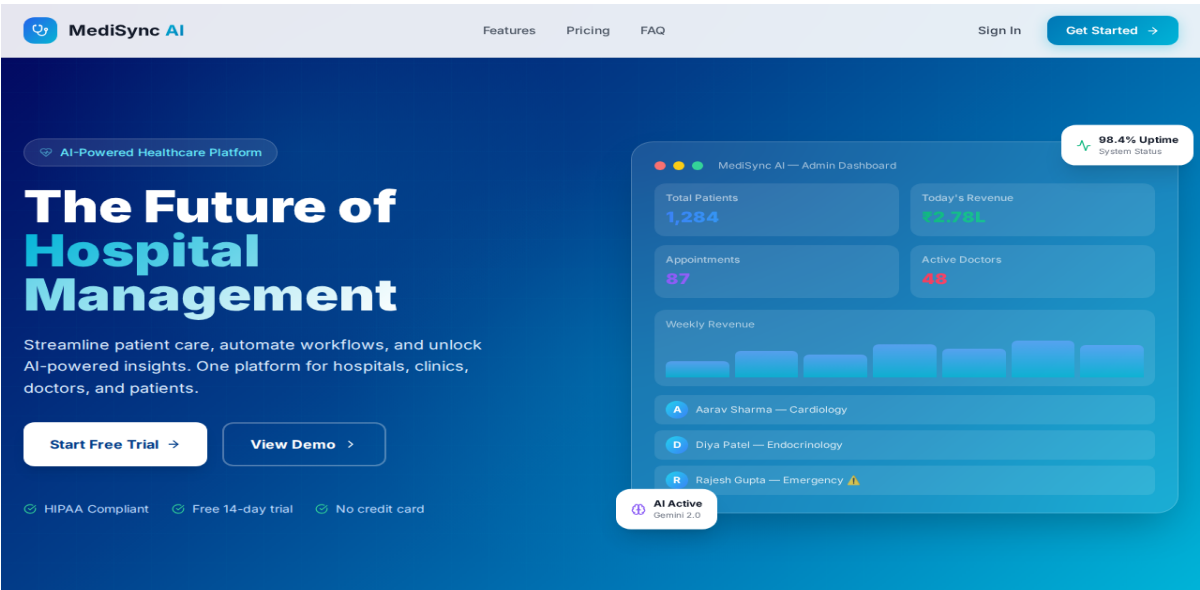


# MediSync AI

# Hospital Management System User Manual & Visual GUI Guide

Step-by-Step Operational Workflows, Role-Based Portals, and GUI Visuals



|                              |                                         |
|------------------------------|-----------------------------------------|
| Version: 1.0.0 (Enterprise)  | System Release: MediSync Web Suite      |
| Audience: All Hospital Roles | Security Level: Confidential & Internal |
| Updated: May 2026            | Platform: Full-Stack React / Node.js    |

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## 1. System Overview & Architecture

**MediSync AI** is a multi-role Hospital Management System designed to optimize clinical workflow, automate front-desk intake, unify e-prescribing, and empower patients through self-service capabilities.

- **Frontend Client:** React 18, TypeScript, Vite, Tailwind CSS, Recharts, Lucide Icons.
- **State Management:** Zustand store with local and API persistence.
- **Backend API:** Node.js, Express, TypeScript, JWT Authorization.
- **Database:** PostgreSQL / SQLite fallback.

## 2. Getting Started & Account Registration

### Creating a New Account:

1. Open `http://localhost:5173/register` in your web browser.
2. Enter your full name, email address, password, and select your role.
3. Click **Create Account** to log in.

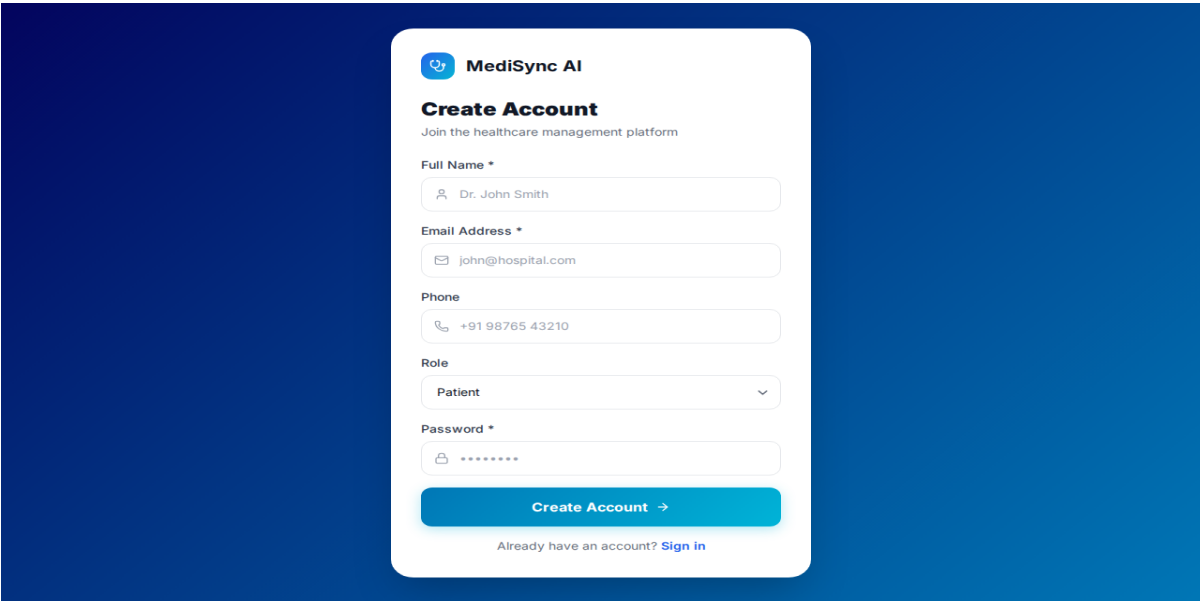


Figure 1: MediSync AI Account Registration GUI Page

Logging In:

Navigate to `/login` and select any Quick Demo Login role badge or enter email/password.

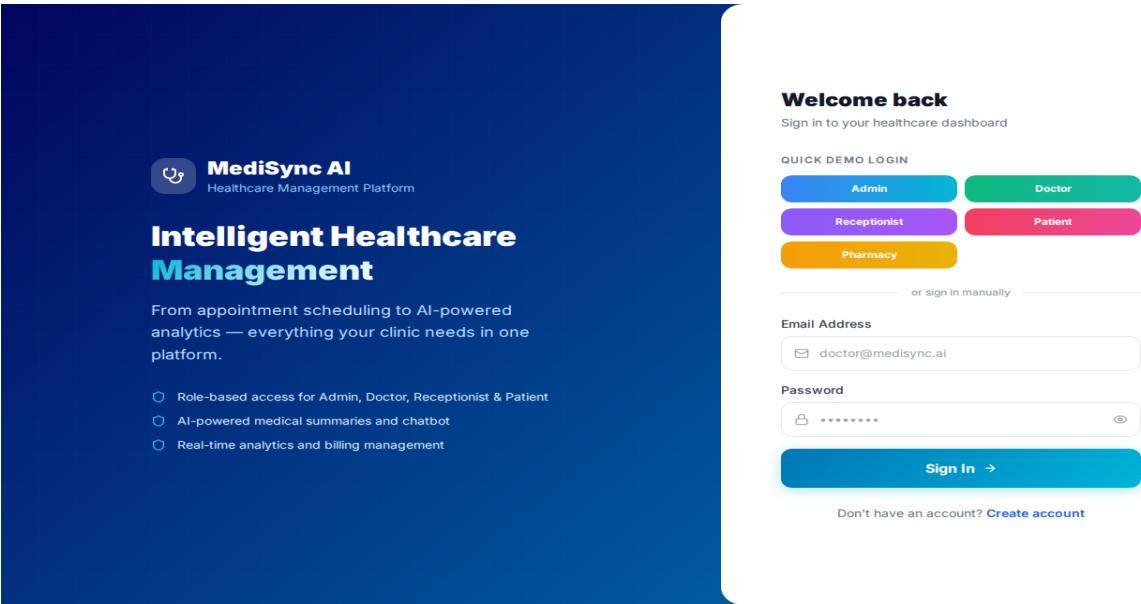


Figure 2: MediSync AI User Login GUI Page

| Role         | Email Address            | Password    |
|--------------|--------------------------|-------------|
| Admin        | admin@medisync.ai        | admin123    |
| Doctor       | doctor@medisync.ai       | doctor123   |
| Receptionist | receptionist@medisync.ai | recept123   |
| Patient      | patient@medisync.ai      | patient123  |
| Pharmacy     | pharmacy@medisync.ai     | pharmacy123 |

### 3. Admin Portal Guide

The Admin Dashboard provides real-time oversight of hospital KPIs, patient directories, doctor scheduling, billing, and analytics.

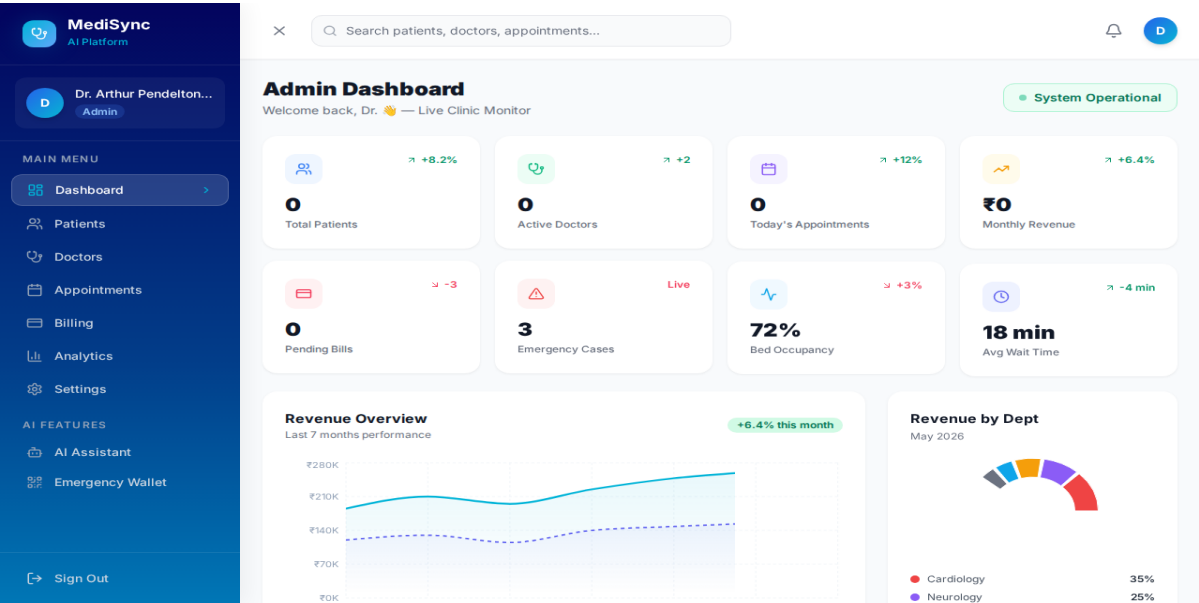


Figure 3: Admin Executive Dashboard Overview GUI

#### Managing Patient Directory:

Click **Patients** on the sidebar (`/dashboard/admin/patients`) to search, filter by condition, or onboard new patient profiles.

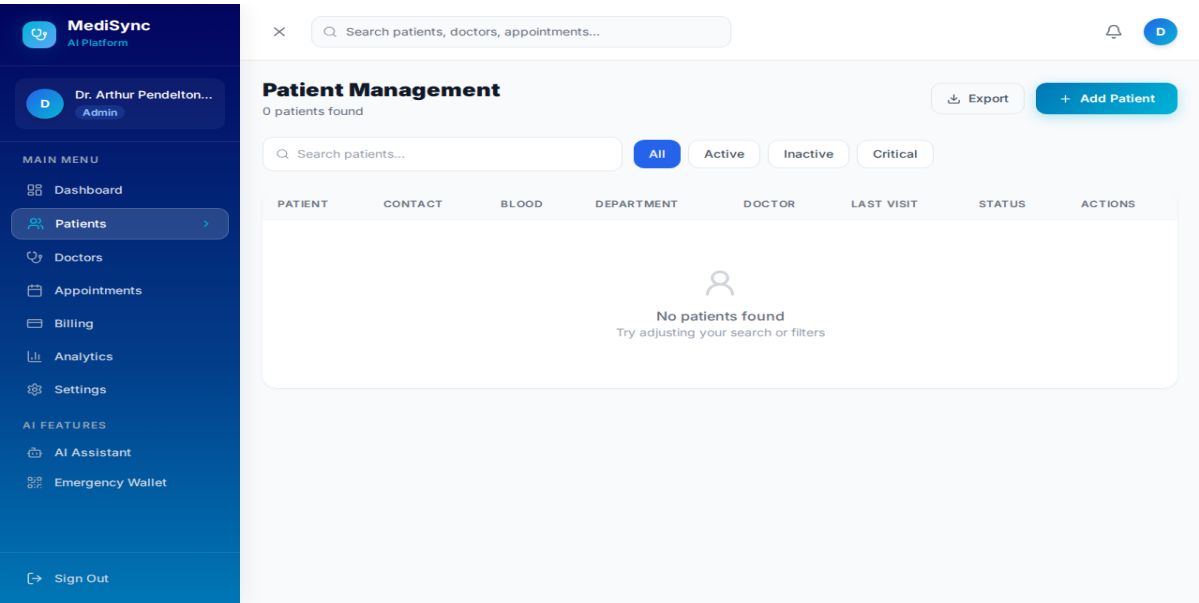


Figure 4: Admin Patient Management & Directory GUI

#### Doctor & Staff Onboarding:

Click **Doctors** on the sidebar (`/dashboard/admin/doctors`) to onboard new physicians and set consultation fees.

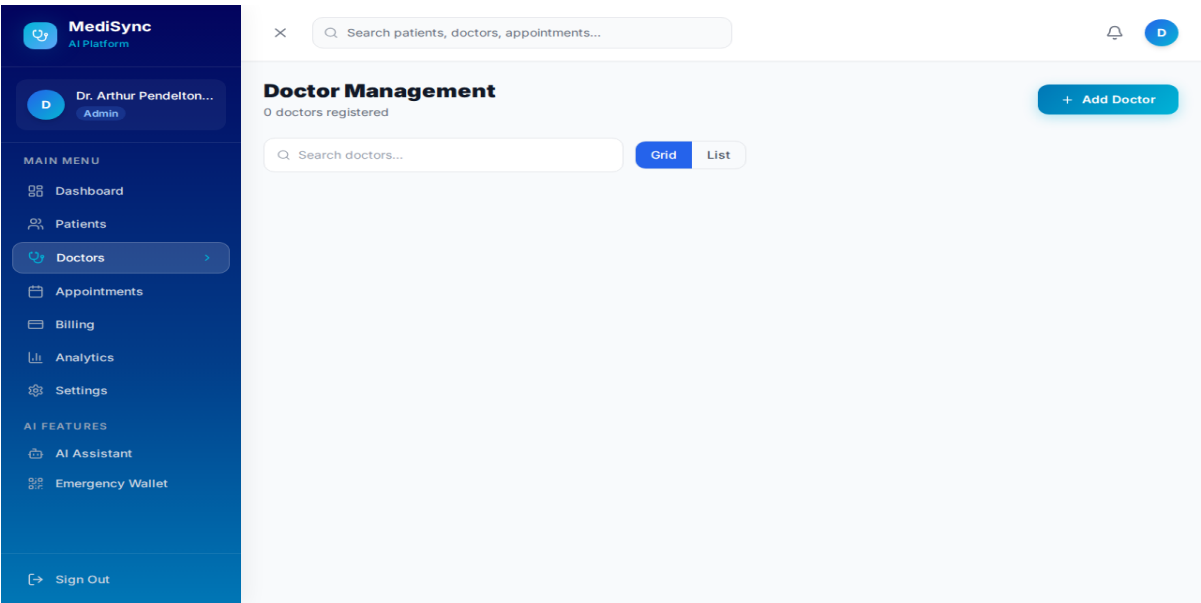


Figure 5: Admin Doctor Directory & Onboarding GUI

Billing Oversight & Invoices:

Navigate to **Billing** (`/dashboard/admin/billing`) to inspect itemized bills and payment states (`Paid`, `Pending`, `Overdue`).

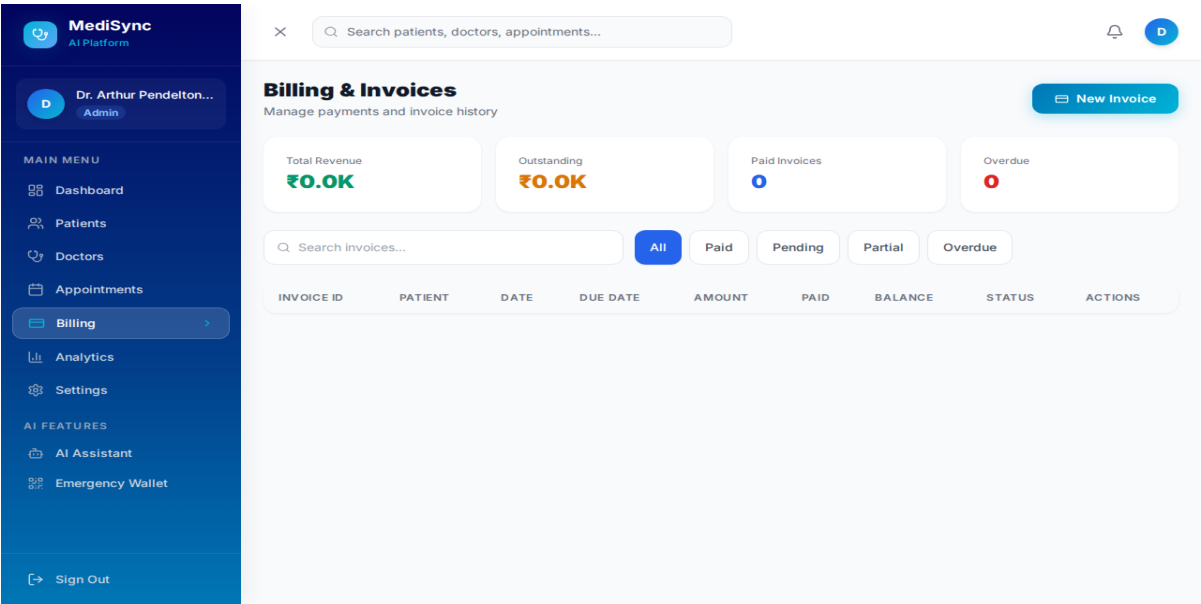


Figure 6: Admin Billing & Financial Invoices GUI

Analytics & Performance Metrics:

Navigate to **Analytics** (`/dashboard/admin/analytics`) to view visual revenue growth trends and bed occupancy rates.

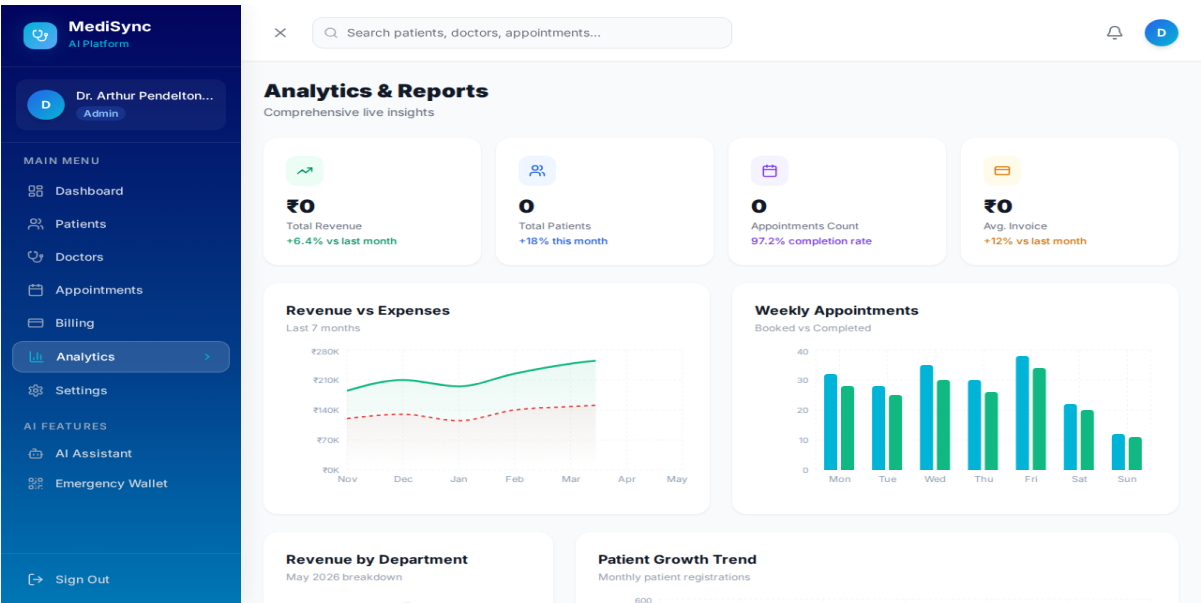


Figure 7: Admin Analytics & Reports Performance Dashboard GUI

System Settings:

Navigate to **Settings** (`/dashboard/admin/settings`) to configure hospital profile parameters and backup options.

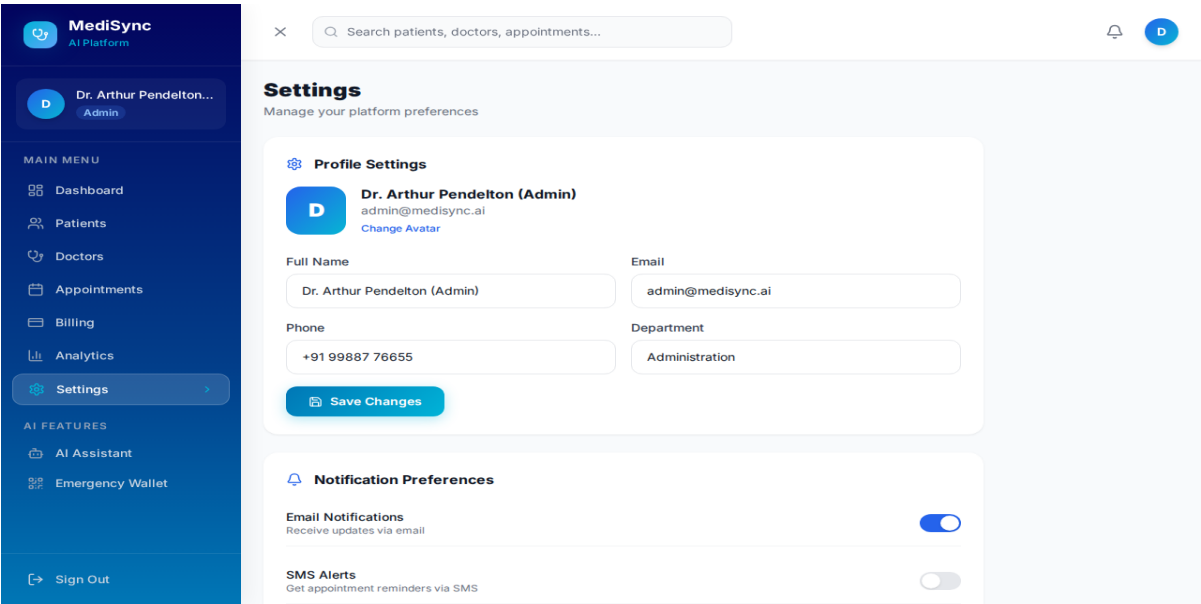


Figure 8: Admin System Settings & Configurations GUI

4. Doctor Portal Guide

Clinicians use this portal to manage patient consultations, review medical histories, and issue signed e-prescriptions.

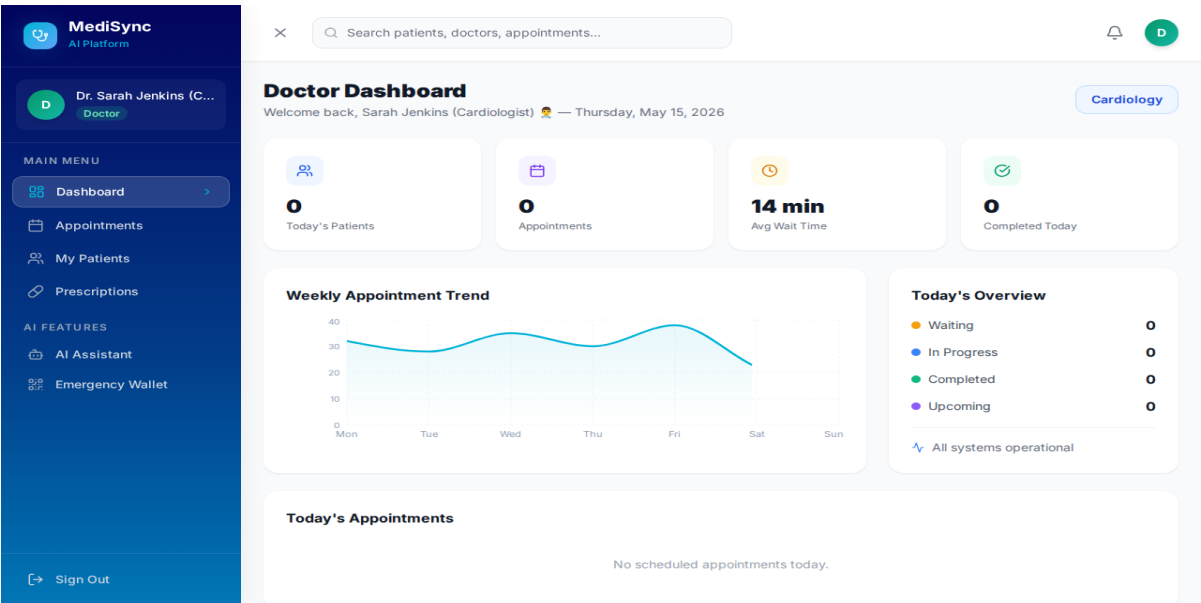


Figure 9: Doctor Consultation Overview Dashboard GUI

Doctor Appointments Desk:

Navigate to **Appointments** (`/dashboard/doctor/appointments`) to update patient lifecycle status (`In Progress`, `Completed`).

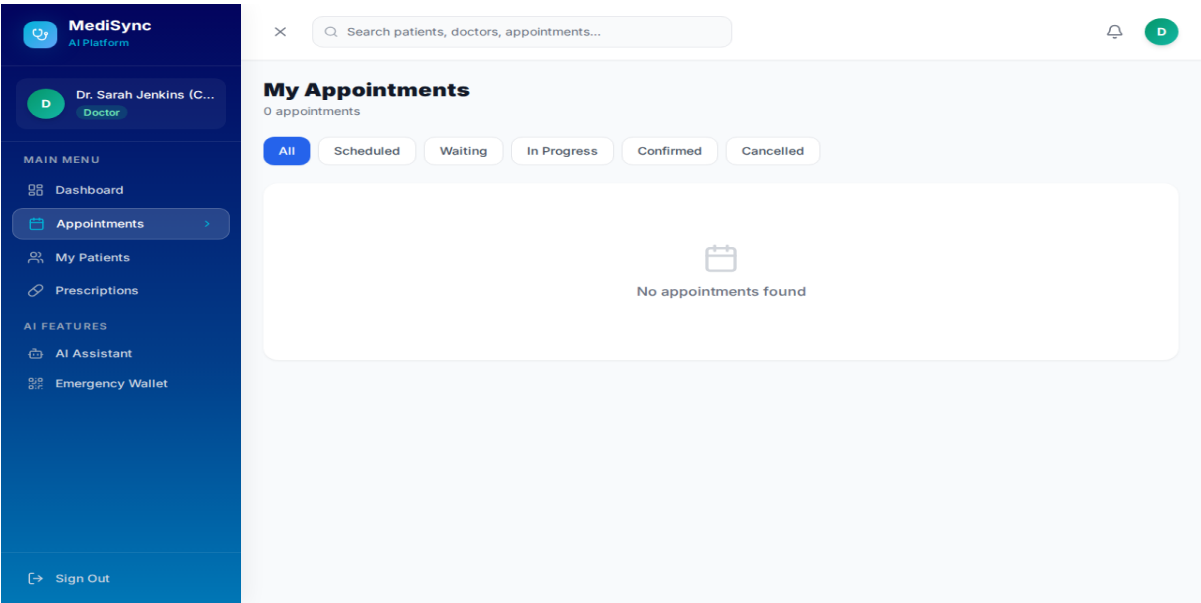


Figure 10: Doctor Consultation Appointments Schedule GUI

Patient EHR Records:

Navigate to **Patients** (`/dashboard/doctor/patients`) to inspect historical vital sign trends and diagnostic notes.

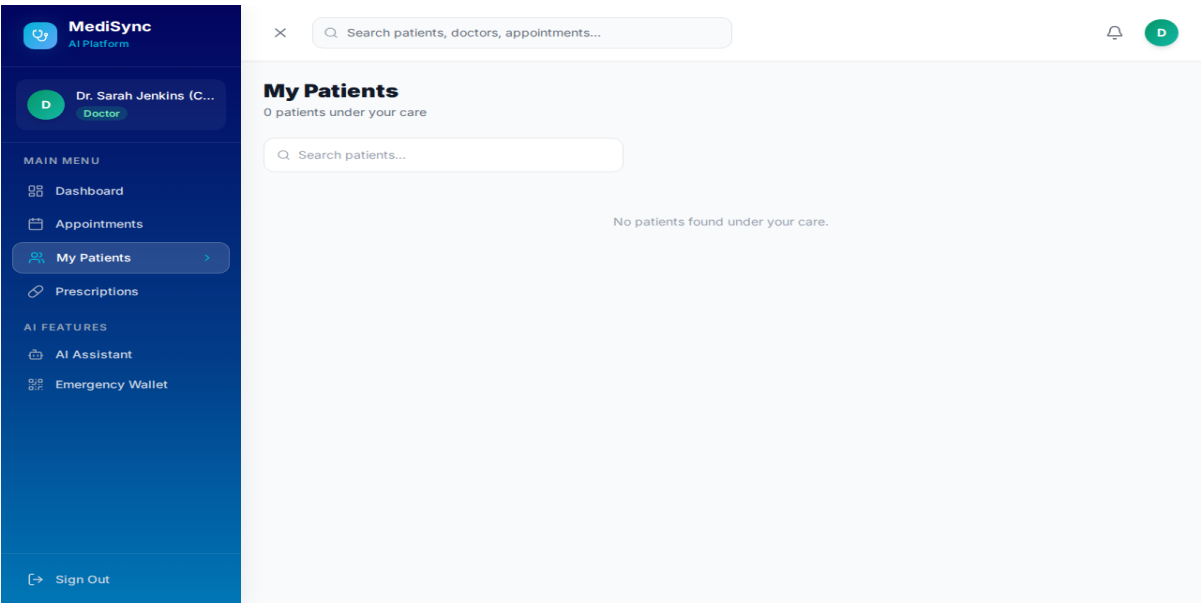


Figure 11: Doctor Patient EHR Health Records GUI

e-Prescribing Workflow:

Navigate to **Prescriptions** (`/dashboard/doctor/prescriptions`). Click **New Prescription**, select patient, specify drug names, dosages, and frequency. Digital signatures are automatically stamped.

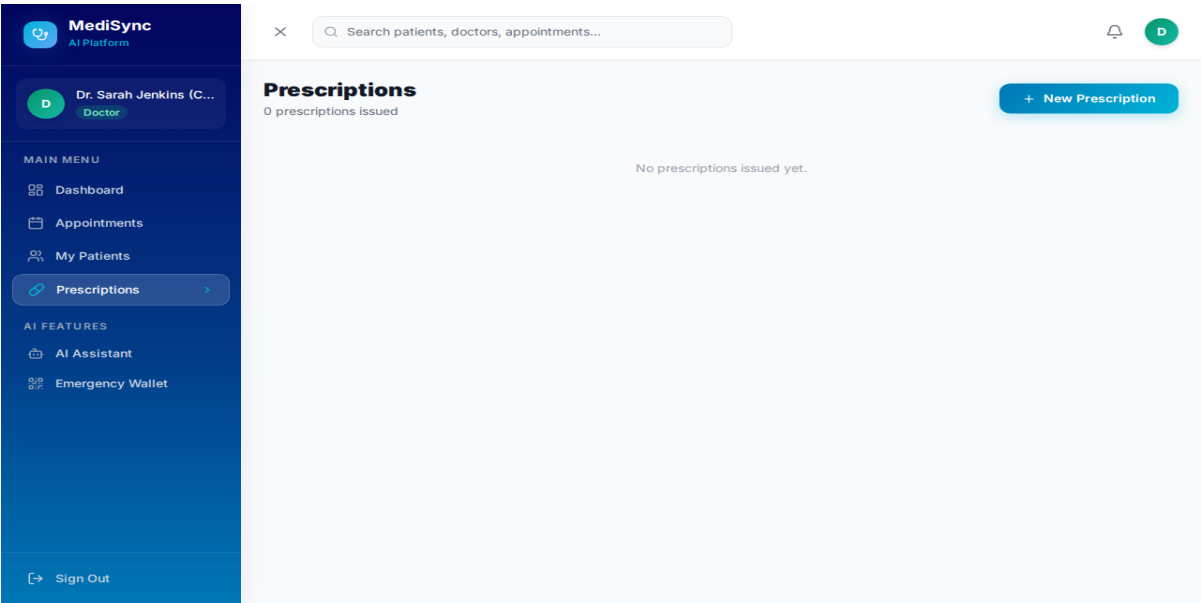


Figure 12: Doctor Electronic Prescribing Interface GUI

5. Receptionist Portal Guide

Designed for front-desk triage, walk-in patient registration, appointment scheduling, and consultation queue optimization.

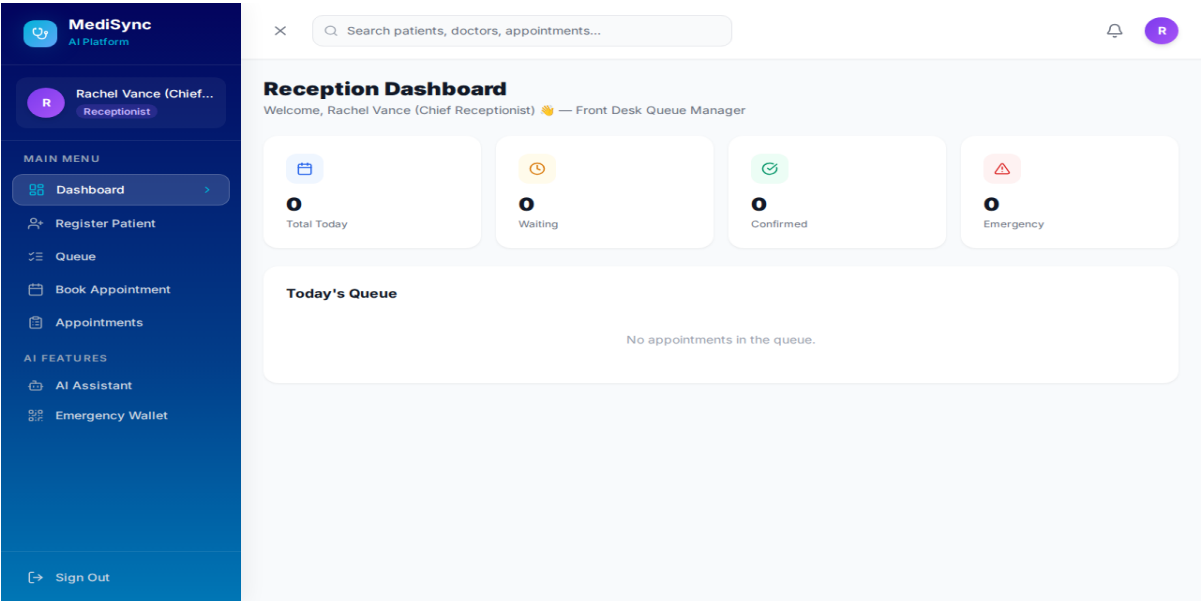


Figure 13: Receptionist Overview Dashboard GUI

Walk-In Patient Registration:

Navigate to **Register Patient** (`/dashboard/receptionist/register-patient`). Fill in vital contact info, blood group, emergency contact, and insurance details.

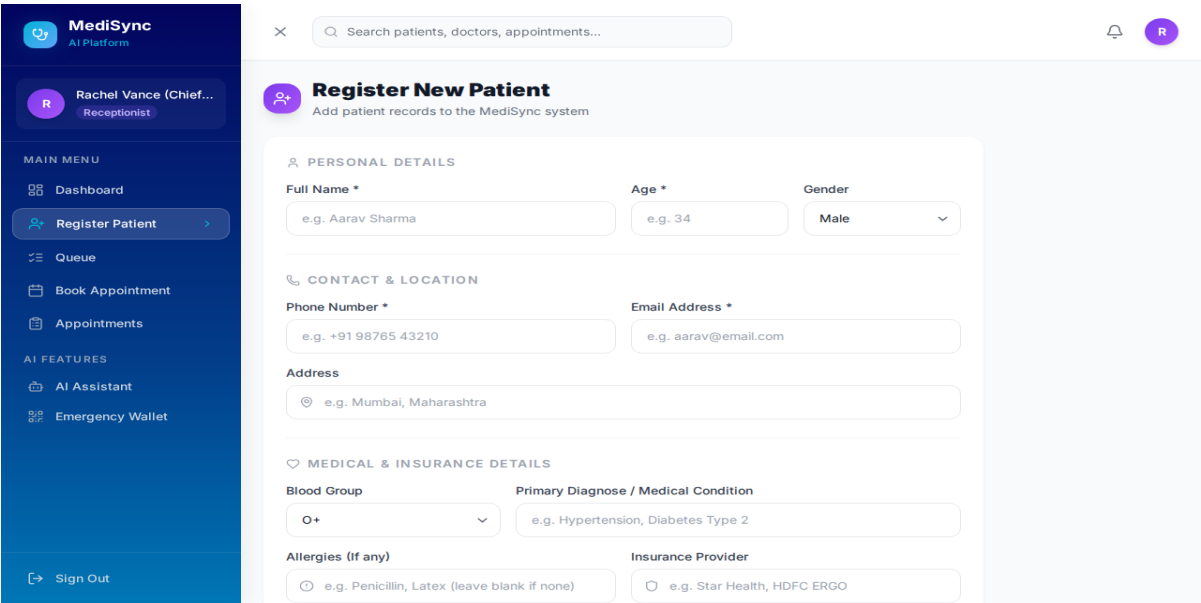


Figure 14: Receptionist Patient Intake Form GUI

Appointment Booking Desk:

Navigate to **Book Appointment** (`/dashboard/receptionist/book`). Select patient, physician, date/time slot, and set priority level.

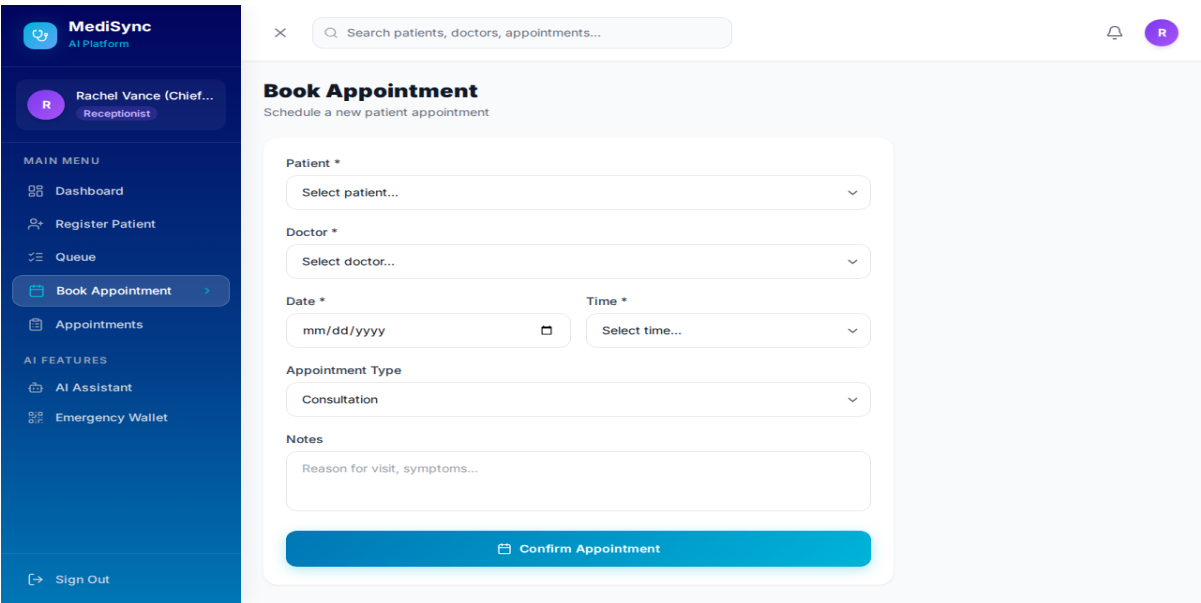


Figure 15: Receptionist Appointment Scheduling Desk GUI

Digital Token Queue Management:

Navigate to **Queue Management** (`/dashboard/receptionist/queue``) to issue sequential consultation tokens and route walk-in patients.

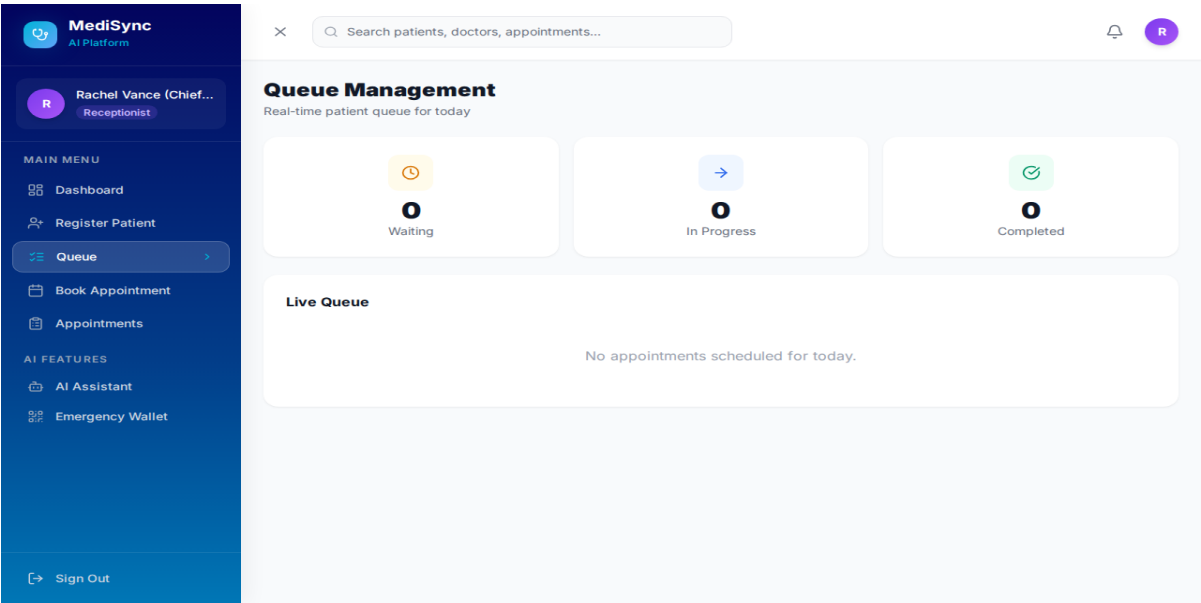


Figure 16: Front Desk Queue & Token Management GUI

6. Patient Portal Guide

Empowers patients to manage their health journey with online booking, medical history access, and bill payments.

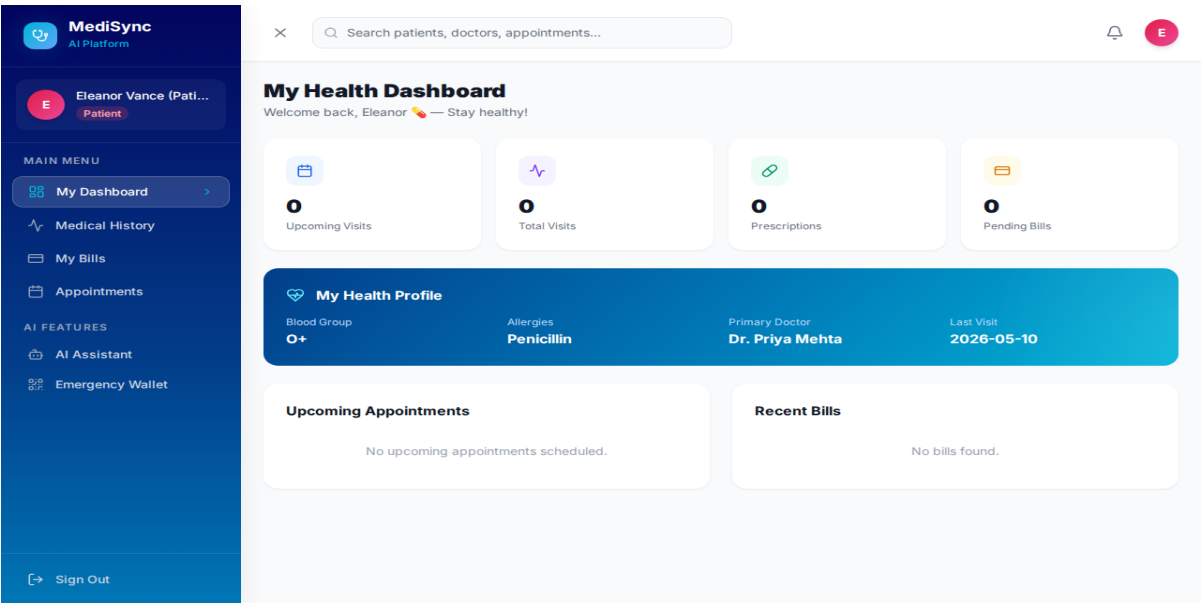


Figure 17: Patient Self-Service Dashboard GUI

Medical History & Prescription Vault:

Navigate to **Medical History** (`/dashboard/patient/history`) to inspect consultation records, past diagnoses, and download signed e-prescriptions.

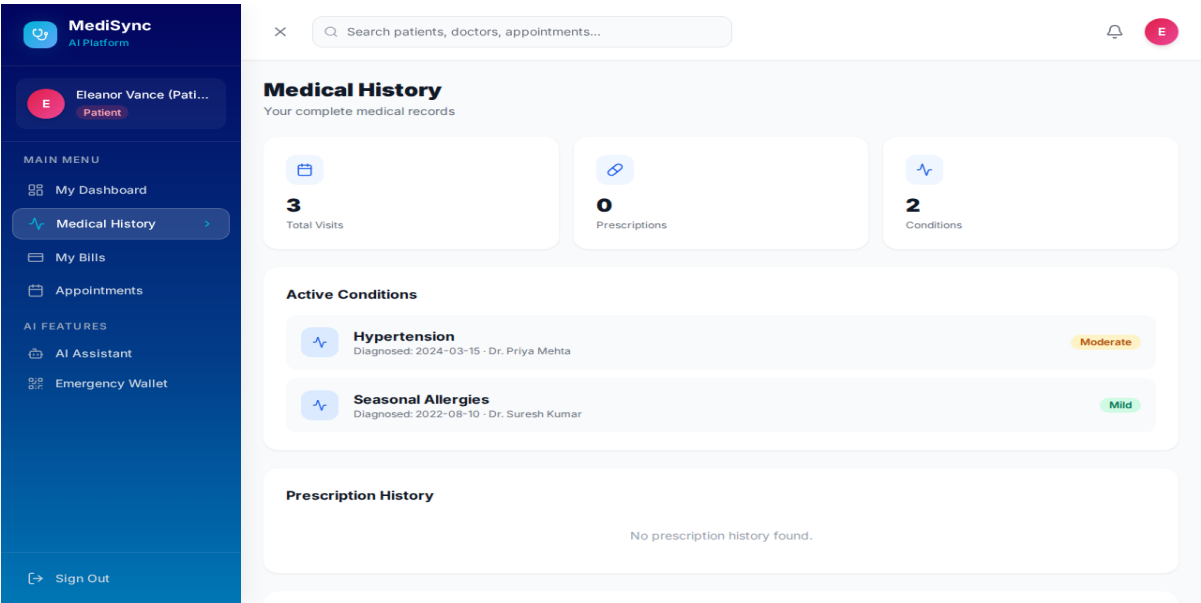


Figure 18: Patient Medical History & Prescription Vault GUI

Digital Bill Payments:

Navigate to **My Bills** (`/dashboard/patient/bills`) to view itemized charges and settle invoices online.

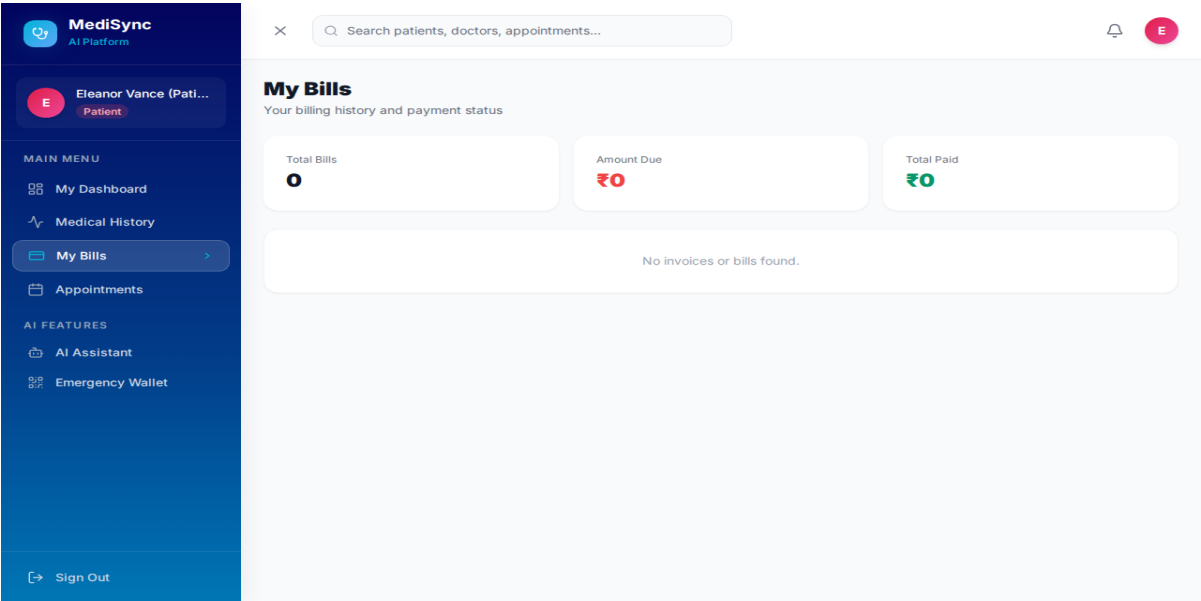


Figure 19: Patient Digital Invoices & Payments GUI

## 7. Pharmacy Portal Guide

Connects doctors with pharmacists for real-time prescription verification and drug dispensing.

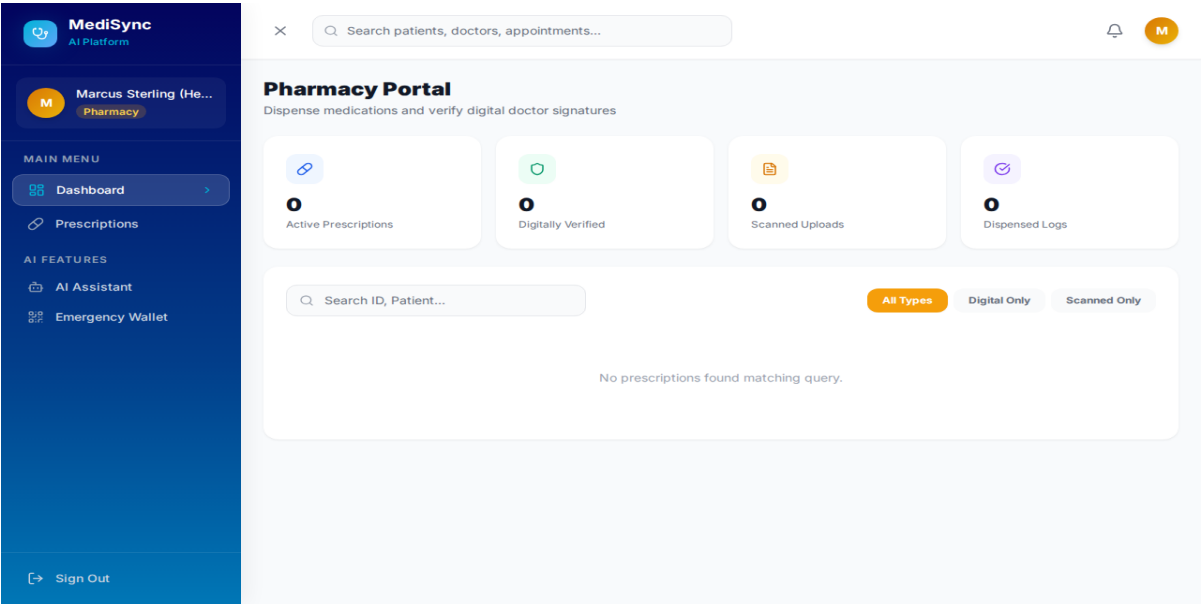


Figure 20: Pharmacy Prescriptions Queue & Dispensing GUI

## 8. Shared AI & Emergency Tools

MediSync AI Clinical Assistant:

Enter patient symptoms to generate intelligent differential diagnosis insights and formatted summaries.

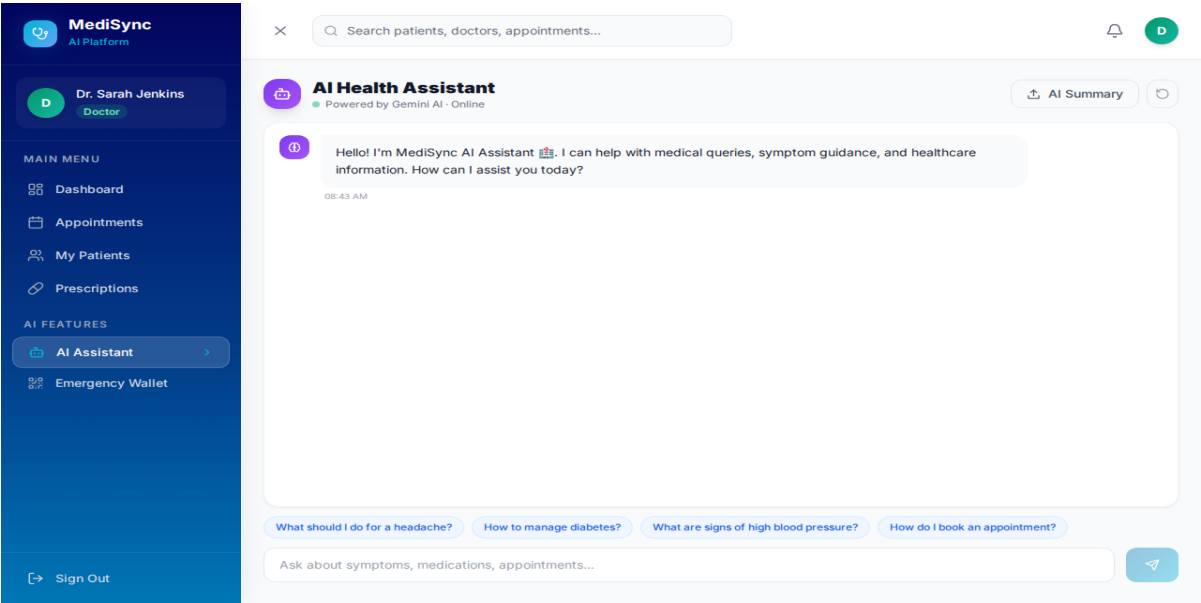


Figure 21: MediSync AI Assistant GUI

Emergency Medical Wallet:

Provides immediate access to critical emergency data: blood group, severe allergies, and emergency contact details.

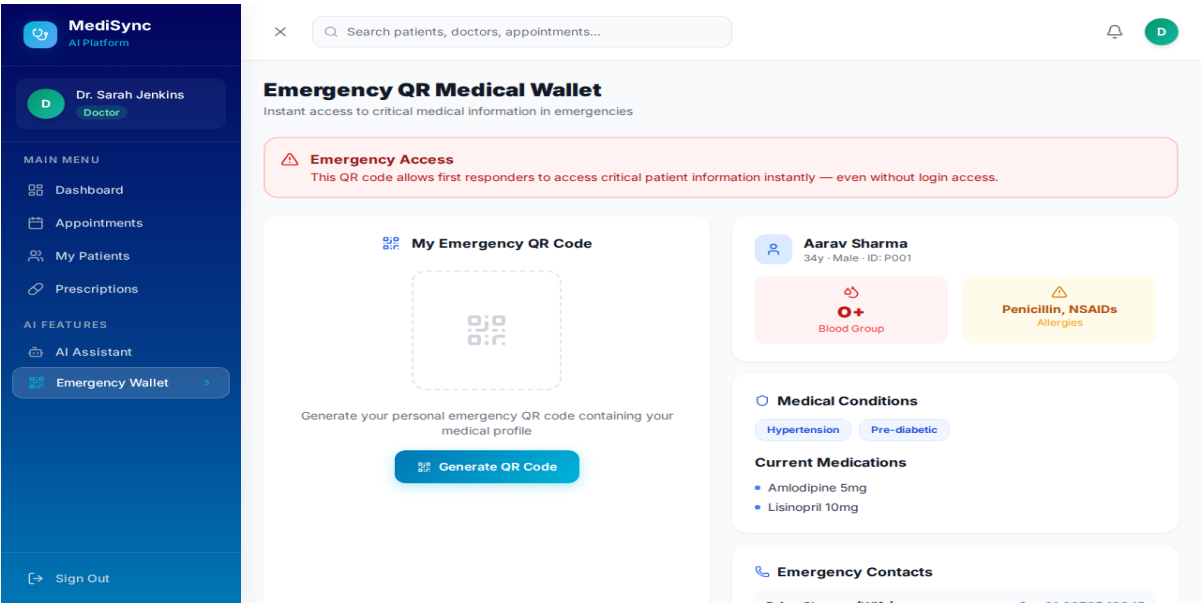


Figure 22: Emergency Medical Wallet GUI

9. API Specifications & Integration

REST API endpoints communicate using JSON format and require JWT Bearer authentication headers.

| Endpoint              | Method     | Description                                        |
|-----------------------|------------|----------------------------------------------------|
| /api/auth/login       | POST       | User authentication & JWT generation               |
| /api/patients         | GET / POST | Fetch patients list or register new patient        |
| /api/doctors          | GET / POST | Fetch doctors directory or add doctor              |
| /api/appointments     | GET / POST | Fetch appointments or schedule new session         |
| /api/prescriptions    | GET / POST | Fetch prescriptions or generate new e-prescription |
| /api/invoices/:id/pay | POST       | Process digital payment for invoice                |

## 10. Troubleshooting & FAQ

**Q: How are emergency appointments prioritized?**

A: Setting appointment priority to **Emergency** immediately places the patient at the top of the doctor queue and broadcasts system-wide notifications.

**Q: Where are digital signatures attached?**

A: Doctors upload their digital signature in Profile Settings. The system stamps this signature automatically onto generated e-prescriptions.